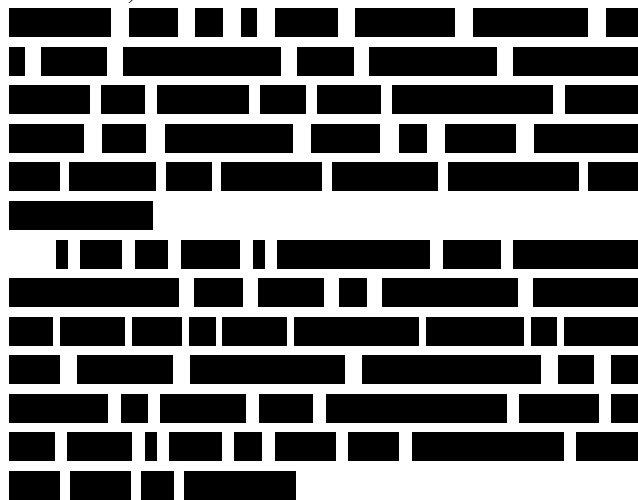


Puzzles to Puzzle You (Remastered)

Welcome to the fourth iteration of *Puzzles to Puzzle You (Remastered)* by Rani Rachavelpula, in which I will discuss the previous puzzle (4D Hedgehog Theorem) and pose the next one.

Thank you all for your patience and understanding. The month of April was a difficult month for *Puzzles to Puzzle You (Remastered)*.



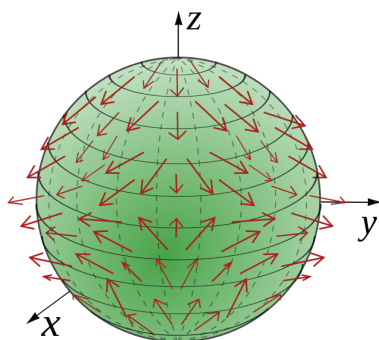
Puzzle #3: 4D Hedgehog Theorem

Imagine a smooth wind on the surface of the Earth always parallel to the ground. It famously follows from the Hedgehog Theorem (also known as the Hairy-Ball Theorem) that the speed of the wind (represented below by arrows) must vanish for at least one point on the Earth's surface.

1. Does this result hold for 4 dimensions?

No

2. Why (bonus)?



First, let us understand what the Hedgehog Theorem says when it is true. Suppose we have a two-dimensional surface S^2 embedded in 3D. Then we can consider all smooth vector fields tangent to S^2 . Then, the Hedgehog Theorem tells us that any smooth vector field on S^2 must vanish at at least one point. There must be a point at which the tangent vector is the 0 vector.

Thus, when we ask whether the Hedgehog Theorem holds in 4D, we are really asking whether there is a nowhere vanishing, smooth vector field tangent to the surface of S^3 .

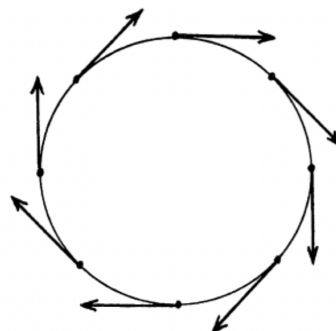
And the answer to that question is no! The Hedgehog Theorem does not hold in 4D! There exists a non-vanishing, smooth vector field tangent to the surface of S^3 . Indeed, the theorem does not hold for S^n for any odd n .

2D case (S^1) One way to see that the theorem may not generalize to all dimensions is to think about the simple case of S^1 . For S^1 , it is clearly possible to define a vector field tangent to its surface.

Define a smooth vector field $V : S^1 \rightarrow \mathbb{R}^2$

$$V(x_1, x_2) = (-x_2, x_1)$$

Thus, we can 'comb' the circle S^1 .



4D case (S^3) The strategy for generating a smooth non-vanishing vector field tangent to S^1 can be extended to S^3 , which shows that the Hedgehog Theorem does not work in 4D. Recall,

$$S^4 = \{x \in \mathbb{R}^4 | x_1^2 + x_2^2 + x_3^2 + x_4^2 = 1\}$$

Define a vector field $V : S^3 \rightarrow \mathbb{R}^4$

$$V(x_1, x_2, x_3, x_4) = (x_2, -x_1, x_4, -x_3)$$

Proposition V defines a non-vanishing vector field on S^3 .

Pf: Clearly, V is a smooth map.

Also, V is non-vanishing. Since $0 \notin S^3$, $0 \notin V(S^3)$.

Finally, $V(x) \perp (x)$ since

$$\begin{aligned} V(x) \cdot x &= (y, -x, w, -z) \cdot (x, y, z, w) \\ &= xy - xy + wz - wz \\ &= 0. \end{aligned}$$

Thus, V is a smooth, non-vanishing vector field tangent to S^3 $\square$.

Remark It's easy to see that whenever S^n has an even number of coordinates, I can use this strategy to generate a smooth, non-vanishing vector field tangent to S^n at every point. Since $S^n \subseteq \mathbb{R}^{n+1}$, this strategy works for S^n where n is odd. Thus, $\mathbb{R}^{n+1}$ is of even dimension. As such, the Hedgehog Theorem works in 3D, but not 4D.

My Strategy When first contemplating this problem, I did not immediately jump to formulating a smooth, non-vanishing vector field tangent to S^3 . First, I had to figure whether or not the theorem seemed true or false for 4D. To this end, I reviewed the proof of the Hedgehog Theorem in 3D. In particular, I looked at the classic proof in Munkres involving homotopy theory. I saw that the proof requires proving the following lemma.

Lemma : For even $n = 2k, k \in \mathbb{N}$, the identity map on $Id_{S^{2k}} : S^{2k} \rightarrow S^{2k}$ is not homotopy equivalent ($\sim$) to the antipodal map $A_{S^{2k}} : S^{2k} \rightarrow S^{2k}$. That is,

$$id_{S^{2k}} \not\sim A_{S^{2k}}$$

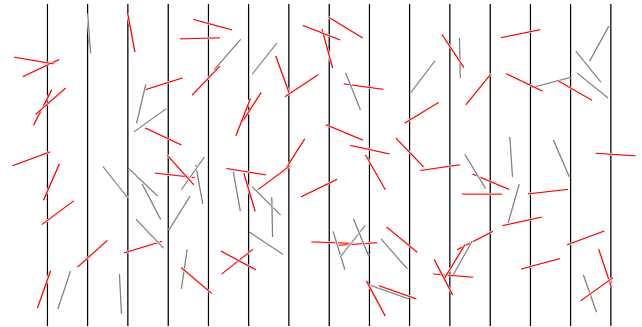
After seeing that this lemma which assumes n is even was necessary for the truth of the Hedgehog Theorem in 3D, I began to think the theorem failed for S^n, n odd.

This is all to say, though my eventual answer seems simple and straightforward, my path to it was definitely not.

Puzzle #4: Buffon's Needle Problem

Buffon's Needle Problem Suppose we have a floor made of parallel strips of wood, each the same width w apart. Suppose further we drop a needle of length l onto the floor, where $l \leq w$.

1. What is the probability that the needle will intersect a line between two strips?



Buffon's needle problem was first posed in the 18th century by Georges-Louis Leclerc, Comte de Buffon. Buffon was an important French naturalist, mathematician, and cosmologist. He has a far-reaching legacy, which includes an asteroid (7420) Buffon named after him.



Buffon's gambling habit naturally led him to consider mathematical questions in geometric probability, the subdomain of mathematics in which the solution to Buffon's Needle Problem is a key theorem.